Supplementary Material for
Valuations, Simplex Embeddings, and Product Rules in Finite Quantum Logics

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# 1 Scope of This Supplement

This supplement records the computational artifacts used for the projector-cone and vector-generated closure tables in the paper. It is designed to make clear which entries are exact-certified and which entries are numerical rationalizations.

The accompanying archive contains:

- `simplex_ultimate_audit.py`, the driver that writes all result manifests and optionally reruns the numerical and exact audits;
- `simplex_projector_cone_vector_audit.py`, the projector-cone audit with rational primal and dual certificate checks;
- `simplex_operational_closure_vector_audit.py`, the vector-generated closure audit;
- `simplex_original_formalism_exact_cdd_v10_completed_audit.py`, the earlier exact-audit script retained for provenance;
- JSON manifests and logs produced by the ultimate driver, especially
  - `projector_cone_audit_run.json`,
  - `operational_closure_audit_run.json`,
  - `exact_certificate_generation_status.json`.

Exact regeneration of rational facets and exact certificates requires a fraction/GMP-enabled `pycddlib`/cddlib installation. A float-only `pycddlib` installation can reproduce numerical LP summaries, but those runs are not exact certificates.

# 2 Certificate Format

Let

$$M_0 = I_E^T G I_\Omega, \qquad M_1 = I_E^T G(D - \mathrm{Id}) I_\Omega,$$

and let $C_{\ell k} = (h_\ell^E)^T h_k^\Omega$. The primal certificate for a candidate value $r_*$ is a rational nonnegative matrix $\sigma$ satisfying

$$\sum_{\ell,k} \sigma_{\ell k} C_{\ell k} - r_* M_1 = M_0.$$

The dual certificate is a rational matrix $Y$ satisfying

$$\langle Y, C_{\ell k}\rangle \le 0 \quad \text{for all } \ell, k, \qquad -\langle Y, M_1\rangle \le 1,$$

and

$$\langle Y, M_0 \rangle = r_*.$$

Together, these identities certify that $r_*$ is the exact optimum for the lower-bounded LP. Since all certified rows satisfy $0 \le r_* \le 1$, the same certificate applies to the implemented LP with the additional bound $r \le 1$.

The generated `supplement_run/exact_projector_certificates/` directory contains one JSON certificate file per exact projector-cone row. Each file contains

$$I_\Omega,\ I_E,\ D,\ H_\Omega,\ H_E,\ M_0,\ M_1,\ r_*,\ \sigma,\ Y,$$

with rational entries serialized as strings.

# 3 Program Commands

The light manifest run is:

```
python simplex_ultimate_audit.py --outdir supplement_run
```

This writes all table entries, certificate-status manifests, dependency reports, and program hashes.

The full numerical regeneration run is:

```
python simplex_ultimate_audit.py --outdir supplement_run --run-full
```

If the local `pycddlib` build is float-only, append `--allow-float-cdd`. Those runs are numerical only.

The exact-certificate run is:

```
python simplex_ultimate_audit.py --outdir supplement_run ^
  --run-full --save-exact-certificates
```

and requires exact fraction/GMP cddlib support.

The overnight run of these three commands produced the files in `supplement_run/`. The projector-cone audit confirms exact rational primal and dual certificates for all six rational projector-cone rows. The closure audit reproduces the closure values with exact cddlib facets, but its JSON status fields record `--certify not requested` for the primal closure rows and "dual numerical solve has no primal sigma certificate" for the Tkadlec cutting-dual rows. Hence the overnight run changes no table value and does not upgrade any closure row to exact-certified status.

# 4 Projector-Cone Results

| Configuration | $d$ | projectors | acc. dim. | facets | $r$ | exact? | status |
|---|---|---|---|---|---|---|---|
| Firefly logic $L_{12}$ | 3 | 5 | 4 | 5 | 0 | yes | exact optimum |
| Completed pentagon $C_5$ | 3 | 10 | 6 | 12 | $\approx 0.119140568134$ | no | numerical/algebraic |
| Specker bug/TIFS, nonfull | 3 | 13 | 6 | 26 | $\approx 0.31645570$ | no | numerical |
| $\Gamma_3$, nonseparating | 3 | 25 | 6 | 230 | $\approx 0.50219773$ | no | numerical |
| Tkadlec Fig. 2, nonunital | 3 | 37 | 6 | 276 | 35/64 | yes | exact optimum |
| Yu–Oh 13-ray projector fragment | 3 | 13 | 6 | 24 | 3/8 | yes | exact optimum |
| Yu–Oh 25-ray orthogonal completion | 3 | 25 | 6 | 96 | 21/44 | yes | exact optimum |
| Cabello–Estebaranz–Garcia-Alcaine 18–9 | 4 | 18 | 10 | 146 | 1/3 | yes | exact optimum |
| Peres–Mermin 24–24 completion | 4 | 24 | 10 | 120 | 4/9 | yes | exact optimum |

# 5 Vector-Generated Closure Results

No row in this table is claimed as an exact-certified optimum. The fractions are rationalizations of floating-point optima from the closure audit.

| Configuration | closure | $d$ | $(\lvert\Omega\rvert, \lvert E\rvert)$ | $(f_\Omega, f_E)$ | solver | $r$ |
|---|---|---|---|---|---|---|
| Firefly logic $L_{12}$ | closed effects | 3 | $(6, 11)$ | $(5, 5)$ | primal | $\approx 0$ |
| Firefly logic $L_{12}$ | closed effects+preparations | 3 | $(11, 11)$ | $(5, 5)$ | primal | $\approx 0$ |
| Yu–Oh 13 | closed effects | 3 | $(14, 51)$ | $(24, 96)$ | primal | $\approx 9/20$ |
| Yu–Oh 13 | closed effects+preparations | 3 | $(51, 51)$ | $(96, 96)$ | primal | $\approx 21/44$ |
| Yu–Oh 25 completion | closed effects | 3 | $(26, 51)$ | $(96, 96)$ | primal | $\approx 21/44$ |
| Yu–Oh 25 completion | closed effects+preparations | 3 | $(51, 51)$ | $(96, 96)$ | primal | $\approx 21/44$ |
| Tkadlec Fig. 2, nonunital | closed effects | 3 | $(38, 123)$ | $(276, 1140)$ | cutting dual | $\approx 445/792$ |
| Tkadlec Fig. 2, nonunital | closed effects+preparations | 3 | $(123, 123)$ | $(1140, 1140)$ | cutting dual | $\approx 192/337$ |
| Cabello 18–9 | closed effects | 4 | $(19, 121)$ | $(146, 120)$ | primal | $\approx 4/9$ |
| Cabello 18–9 | closed effects+preparations | 4 | $(121, 121)$ | $(120, 120)$ | primal | $\approx 4/9$ |
| Peres–Mermin/Peres 24 | closed effects | 4 | $(25, 139)$ | $(120, 120)$ | primal | $\approx 4/9$ |
| Peres–Mermin/Peres 24 | closed effects+preparations | 4 | $(139, 139)$ | $(120, 120)$ | primal | $\approx 4/9$ |

# 6 Input Coordinates

The rational ray coordinates are stored directly in `simplex_projector_cone_vector_audit.py` under these names:

`fireflyL12`, `yuOh13`, `yuOh25`, `tkadlecFig2Nonunital`, `cabello18`, `peresMermin24`.

The Yu–Oh 25 list is generated from the Yu–Oh 13 list by adding the primitive cross-product completion of every orthogonal pair. The algebraic pentagon branch is generated by `kcbs_completed_pentagon_rays_float()`.

# 7 Numerical Status

The projector-cone rows marked "exact optimum" have both rational primal feasibility and rational dual optimality certificates. These are serialized in the `exact_projector_certificates/` directory of the run output. The completed pentagon, Specker bug, and $\Gamma_3$ rows remain numerical evidence in the present archive. All vector-closure rows remain numerical in the overnight run: the rational closure facets were exact, but exact primal–dual closure certificates were not produced.